Case 1

1) YTM is an annualized internal rate of return (IRR) only if three conditions are met: hold to maturity, no default (all coupon and principal payments in full and exactly on time), and she can reinvest every coupon payment at the exact same 6% annual rate.

2)

$N = 2(2032 - 2026) + 1 = 13, c = 9\frac{\%}{2} = 4.5\%, i = 7.6\frac{\%}{2} = 3.8\%, f = \frac{3}{180}, P = 100, A = \frac{1-(1+i)^{-N}}{i}$

$\text{FP} = (1+i)^f(P + P(c-i)A) = 107.14$

3) $\text{AI} = fC = fPc = 0.075, \text{CP} = \text{FP} - \text{AI} = 107.07$

Quoted price (also called clean price) is the value of full price excluding the accrued interest. The difference between the quoted price and the full price on the settlement date is the accrued interest. Dealers quote the clean price so that market price fluctuations aren't distorted by how close the bond it to its next interest payment. However, Mrs. Smith will actullay pay the full price because buyer must compensate the seller for accrued interest.

4) By $\left(1 + \frac{\text{APR}_{12}}{12}\right)^{12} = \left(1 + \frac{\text{APR}_2}{2}\right)^2$, we have $\text{APR}_2 = 8.76\%$, which higher than Heavy Equipment Bond's 7.6%. Thus, the Rock Bond is more attractive."

Case 2

1) $N = 20, c = 6.5\frac{\%}{2} = 3.25\%, P = 100, i = 9\frac{\%}{2} = 4.5\%, f = 0, A = \frac{1-(1+i)^{-N}}{i}$

$\text{FP} = (1+i)^f(P + P(c-i)A) = 83.74$

2) Since 6.5% coupon rate is less than 9% market discount rate (YTM), investors demand a price reduction to compensate for lower interest payment, making it discount bond.

3)

- enter the calculated bond purchase price ($83.74) as the opening carrying value at period 0.
- for each period, calculate the true economic interest expense by multiplying the opening carrying value by the semi-annual market discount rate (4.5%).
- subtract the fixed cash coupon payment ($3.25) from the interest expense. This difference represents the unearned discount that must be added back to the bond's book value.
- repeat 20 times updating the carrying value by adding the period's discount amortization. By period 20, the carrying value will land on the$100 par value.

Case 3

1) By $P_i = \frac{C}{2}\left(\sum_1^{i-1} d_{0.5i}\right) + \left(100 + \frac{C}{2}\right)d_{0.5i}$ and $d_t = \frac{1}{(1+\frac{z_t}{2})^t}$, we have

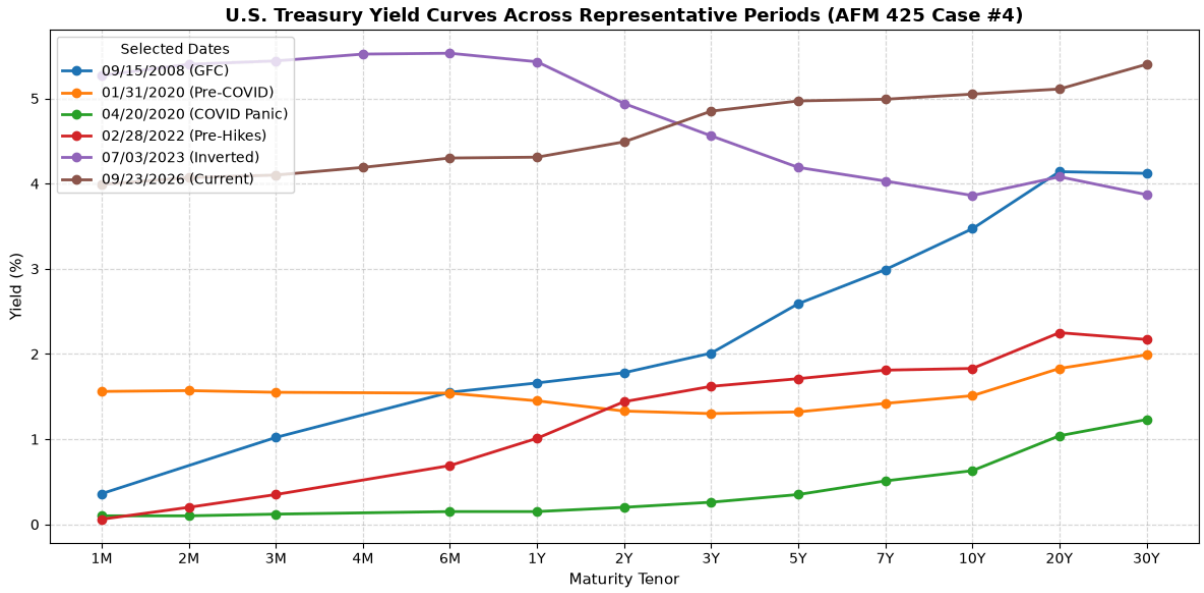
- t=0.5 Y (1 period)
  $P_1 = (100 + 1.875)d_{0.5} = 99.177975 \to d_{0.5} = 0.973526. z_{0.5} = 2\left(\frac{1}{d_{0.5}} - 1\right) = 5.4387\%.$
- t=1 Y (2 period)
  $P_2 = 2.525d_{0.5} + 102.525d_1 = 99.399571 \to d_1 = 0.945539. z_1 = 2\left(d_1^{-\frac{1}{2}} - 1\right) = 5.6791\%.$
- t=1.5 Y (3 period)
  $P_3 = 3.195(d_{0.5} + d_1) + 103.195d_{1.5} = 100.547780 \to d_{1.5} = 0.914932. z_{1.5} = 2\left(d_{1.5}^{-\frac{1}{3}} - 1\right) = 6.0195\%.$
- t=2 Y (4 period)
  $P_4 = 1.22(d_{0.5} + d_1 + d_{1.5}) + 101.22d_2 = 94.424926 \to d_2 = 0.898710. z_{2.0} = 2\left(d_2^{-\frac{1}{4}} - 1\right) = 5.4107\%$
- t=2.5 Y (5 period) $P_5 = 3.65(d_{0.5} + ... + d_2) + 103.65d_{2.5} = 105.038305 \to d_{2.5} = 0.881948. z_{2.5} = 2\left(d_{2.5}^{-\frac{1}{5}} - 1\right) = 5.09\%.$

- t=3 Y (6 period) $P_6 = 3.235(d_{0.5} + ... + d_{2.5}) + 103.235d_3 = 104.589915 \rightarrow d_3 = 0.868518. z_3 = 2\left(d_3^{-\frac{1}{6}} - 1\right) = 4.7610\%$

2) Cash flow happens at t=1,1.5,...,3 from now on and each pay coupon $\frac{C}{2}$. We need to discount P_0 forward by $d_{0.5}$ to get $P_{0.5}$, thus $P_{0.5} = \frac{2.5(d_1+d_{1.5}+d_2+d_{2.5})+102.5d_3}{d_{0.5}} = 100.7943$.
$100.7942 = \sum_{k=1}^{5} \frac{2.5}{(1+r)^k} + \frac{100}{(1+r)^5} \rightarrow r \approx 2.3308\%$. Thus, YTM $= 2(2.3308\%) = 4.66\%$.
3) By $P = \sum_{k=1}^{6} \frac{\text{CF}_k}{\left(1+\frac{z_k}{2}+0.0075\right)^k}$, we have $P = 99.21$.
4) $95.45 = \sum_{k=1}^{5} \frac{2.675}{(1+r)^k} + \frac{100}{(1+r)^6} \rightarrow r = 3.5298\%$. So YTM $= 2r = 7.06\%$
Nominal Spread = YTM − Benchmark Yield = 7.06% − 4.50% = 256 bps.
$95.45 = \sum_{k=1}^{6} \frac{2.675}{\left(1+\frac{z_k}{2}+s\right)^k} + \frac{100}{\left(1+\frac{z_6}{2}+s\right)^6}$
By excel's goal seek function, we have s=1.1259%. Z=2 s=2.2518%=225.18 bps.

Case 4

1) Representative dates I pick for the respective periods are: 2008/9/15, 2020/1/31, 2020/4/20, 2022/2/28, 2023/7/3, 2026/9/23.
Yield curves plot:



a. GFC (2008): Short rates collapsed 1.02% due to emergency Fed cuts and flight-to-safety, while 10Y rates stayed higher (3.47%) expecting eventual economic recovery.
b. Pre-COVID (2020): Rates compressed near 1.30%-1.57%, inverting the 10Y-3M spread as late-cycle momentum slowed.
c. COVID Panic (2020): Emergency rate cuts 0.00%-0.25% and massive QE pushed the entire curve down near historical lows 10Y at 0.63%.
d. Pre-Hikes (2022): Front-end yields remained anchored near zero, but intermediate/long yields rose 10Y at 1.83% pricing in upcoming anti-inflation hikes.
e. Deep Inversion (2023): Aggressive Fed tightening raised short rates to 5.44%, while long rates remained lower 3.86% expecting a policy-induced economic slowdown and eventual rate cuts.
f. Current Period (2026): The curve un-inverted and re-steepened into a normal slope (10Y-2Y = +0.56%) as front-end rates eased while long-end yields rose due to ongoing Treasury issuance.
2) When the Fed hikes short rates to restrictive levels, markets anticipate an economic contraction

that will force future rate cuts. Investors rush to lock in long-term bonds, bidding prices up and driving long-term yields below short-term rates.
Inversion squeezes bank Net Interest Margins (NIM) by making short-term borrowing costlier than long-term lending returns. Banks restrict credit expansion, suppressing business investment and consumer spending.
Bull Steepening: Short rates fall rapidly as the central bank cuts rates to cushion a slowing economy.
Bear Steepening: Long rates rise faster than short rates, driven by high inflation, term premium expansion, or heavy government debt issuance.
Historically, recessions begin not during the deep inversion, but as the curve un-inverts and re-steepens, because lagging credit tightening impacts finally hit corporate refinancings and employment. The recent re-steepening reflects both policy recalibration on the front end and elevated fiscal debt issuance pushing up long-end yields 5.00%+, indicating that macroeconomic risk remains active during this transition window.